DAY 4

SPOT 1

1. Haldane effect
2. Lungs
3. Reduced (deoxygenated) hemoglobin binds more CO_2 and H^+, facilitating CO_2 loading in tissues and unloading in lungs.
4. As bicarbonate (HCO_3^-), dissolved CO_2, or carbaminohemoglobin (any one correct).

SPOT 2

1. Increased H^+ concentration in cerebrospinal fluid (due to rise in arterial PCO_2).
2. Carotid bodies (also aortic bodies, but carotid are more sensitive).
3. CO_2 diffuses across the blood-brain barrier → forms carbonic acid → dissociates to H^+ → stimulates central chemoreceptors → increases firing of inspiratory neurons → increased ventilation.
4. Chronic hypercapnia (e.g., in COPD) or hypoxemia at high altitude — when central chemoreceptors adapt and hypoxia becomes the main drive.

SPOT 3

1. Cyanosis
2. ≥ 5 g/dL of deoxygenated hemoglobin in capillary blood
3. In anemia, total hemoglobin is reduced, so even if oxygen saturation is low, the absolute amount of deoxygenated Hb is insufficient to produce visible bluish color
4. Methemoglobinemia or polycythemia (excess Hb leads to more deoxygenated fraction even at normal PO_2)

SPOT 4

1. Forced Expiratory Spirogram (FEV_1/FVC test)
2. Curve B (flattened, delayed expiration → obstructive pattern)
3. FEV_1 decreases more than FVC → ratio < 70% (reduced)
4. Chronic bronchitis, asthma, or emphysema

SPOT 5

1. Effect of increased ambient pressure during deep-sea diving (hyperbaric exposure)
2. Nitrogen (N_2) dissolves in plasma and tissues due to high pressure

3. Rapid ascent causes dissolved N_2 to come out of solution as bubbles

4. Decompression sickness (“the bends”) — joint pain, dyspnea, neurological symptoms